

# Optimal Enrolment

CITS3001 Advanced Algorithms — Time limit: 3s

Karen is planning her degree and would like to get as much out of the coming year as she can. There are  $N$  units on offer. Unit  $i$  demands  $h_i$  hours of effort, and Karen estimates that if she enrolls in it she will pass it with probability  $p_i/100$ .

Karen has  $H$  hours of effort to spend over the year, and she must commit the full  $h_i$  hours to unit  $i$  to have any chance of passing it. She may enrol in any subset of the units whose total effort is at most  $H$ . The *expected* number of units she passes is the sum of the pass probabilities of the units she enrolls in. Help her choose the subset that maximises it.

## Input

The first line contains two integers  $N$  and  $H$  ( $1 \leq N \leq 100$ ,  $1 \leq H \leq 1000$ ): the number of units on offer and the number of hours Karen has available. Each of the next  $N$  lines contains two integers  $h_i$  and  $p_i$  ( $1 \leq h_i \leq 99$ ,  $0 \leq p_i \leq 100$ ): the hours unit  $i$  requires, and its percentage chance of being passed.

## Output

Output the maximum expected number of units Karen can pass, to exactly two decimal places.

### Sample Input 1

```
3 5
2 60
3 70
4 80
```

### Sample Output 1

```
1.30
```

### Sample Input 2

```
4 10
6 90
5 50
5 50
1 5
```

### Sample Output 2

```
1.00
```